

Problem 1

Let (M, d) be a metric space and $C(M)$ the space of continuous functions $f : M \rightarrow \mathbb{R}$ with metric

$$D(f, g) = \sup_{x \in M} |f(x) - g(x)|.$$

We say that a set of functions $\mathcal{F} \subset C(M)$ is pointwise equicontinuous if for every $x \in M$ and $\varepsilon > 0$ there is a δ (depending on x and $\varepsilon > 0$) for which

$$|f(x) - f(y)| < \varepsilon$$

for all $f \in \mathcal{F}$ and y satisfying $d(x, y) < \delta$. If M is compact show that a pointwise equicontinuous family of functions is a equicontinuous.

Proof. Fix $\varepsilon > 0$ and for each $y \in M$, take a $\delta_y > 0$ for which $|f(x) - f(y)| < \varepsilon/2$ for all $d(x, y) < \delta_y$ and $f \in \mathcal{F}$. The balls $B_y(\delta_y/2)$ for $y \in M$ form an open cover of the compact M , so there is a finite subcover $\{B_{y_i}(\delta_{y_i}/2)\}$ for $y_1, \dots, y_n \in M$. Let $\delta = \min\{\delta_{y_1}/2, \dots, \delta_{y_n}/2\} > 0$ and consider any $x, y \in M$ with $d(x, y) < \delta$. $x \in B_{y_i}(\delta_{y_i}/2)$ for some i because the balls $B_{y_i}(\delta_{y_i}/2)$ cover M , and we note that

$$d(y, y_i) \leq d(x, y) + d(x, y_i) < \delta_{y_i}$$

so $y \in B_{y_i}(\delta_{y_i})$ as well. We then have for all $f \in \mathcal{F}$,

$$|f(x) - f(y)| \leq |f(x) - f(y_i)| + |f(y_i) - f(y)| < \varepsilon$$

because $d(x, y_i) < \delta_{y_i}$ and $d(y, y_i) < \delta_{y_i}$. Hence $\mathcal{F}$ is equicontinuous. □

Problem 2

Let $\{(X_i, d_i)\}_{i=1}^{\infty}$ be a sequence of metric spaces. Let $X = \prod_{i=1}^{\infty} X_i$ and for any $x = (x_i)_i, y = (y_i)_i \in X$ define,

$$d(x, y) := \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{d_j(x_j, y_j)}{1 + d_j(x_j, y_j)}$$

- (a) Show that (X, d) is a metric space.
- (b) Let $x^{(n)} \in X$ and $x \in X$. Show that $x^{(n)} \rightarrow x$ in (X, d) iff $x_i^{(n)} \rightarrow x_i$ in (X_i, d_i) for all i .
- (c) Consider the case $X_i = \mathbb{R}$ and $d_i(x, y) := |x - y|$ for all i . Then $\ell^p \subset X$ for all $1 \leq p \leq \infty$. Compare convergence in ℓ^p with convergence in (X, d) .

Proof. (a). Firstly, we note that the general term of the series $d(x, y)$ is positive, so its convergence follows from the series being bounded.

$$\sum_{j=1}^{\infty} \frac{1}{2^j} \frac{d_j(x_j, y_j)}{1 + d_j(x_j, y_j)} \leq \sum_{j=1}^{\infty} \frac{1}{2^j} = 1$$

hence $d(x, y)$ is well-defined. We also see that $d(x, y) \geq 0$ with equality if and only if each general term of the series is 0. That in turn is equivalent to $d_j(x_j, y_j) = 0$ for all j , and that is equivalent to $x = y$. $d(x, y) = d(y, x)$ follows by symmetry of each d_j .

For the triangle inequality, we note that the function $f : [0, \infty) \rightarrow [0, \infty), x \mapsto x/(1+x)$ has derivative $f'(x) = (1+x)^{-2} > 0$ for all x in its domain. This means that f is increasing, so for all $x, y, z \in X, j$,

$$\frac{d_j(x_j, y_j)}{1 + d_j(x_j, y_j)} = f(d_j(x_j, y_j)) \leq f(d_j(x_j, z_j)) + f(d_j(y_j, z_j)) = \frac{d_j(x_j, z_j)}{1 + d_j(x_j, z_j)} + \frac{d_j(y_j, z_j)}{1 + d_j(y_j, z_j)}$$

because $d_j(x_j, y_j) \leq d_j(x_j, z_j) + d_j(y_j, z_j)$. It then follows that

$$d(x, y) = \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{d_j(x_j, y_j)}{1 + d_j(x_j, y_j)} \leq \sum_{j=1}^{\infty} \frac{1}{2^j} \left(\frac{d_j(x_j, z_j)}{1 + d_j(x_j, z_j)} + \frac{d_j(y_j, z_j)}{1 + d_j(y_j, z_j)} \right) = d(x, z) + d(y, z)$$

so d is a metric.

(b). ($\implies$). Suppose $x^{(n)} \rightarrow x$ in (X, d) and fix $i, \varepsilon > 0$. We can take $N > 0$ such that $d(x^{(n)}, x) < \varepsilon/2^{i+1}$ and $d(x^{(n)}, x) < 1$ for $n \geq N$, and for all such n , we get

$$d_i(x_i^{(n)}, x_i) \leq \frac{2d_i(x_i^{(n)}, x_i)}{1 + d_i(x_i^{(n)}, x_i)} \leq 2^{i+1} \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{d_j(x_j, y_j)}{1 + d_j(x_j, y_j)} < 2^{i+1} \cdot \varepsilon/2^{i+1} = \varepsilon$$

Thus $x_i^{(n)} \rightarrow x_i$ in (X_i, d_i) .

($\impliedby$). Suppose $x_i^{(n)} \rightarrow x_i$ in (X_i, d_i) for each i . Fix $\varepsilon > 0$ and take $k > 0$ for which $1/2^k < \varepsilon/2$. For each $1 \leq i \leq k$ we can take $N_i > 0$ such that $d_i(x_i^{(n)}, x_i) < \varepsilon/2$ for all $n \geq N_i$, and we let $N = \max\{N_1, \dots, N_k\}$. Then for all $n \geq N$, we get

$$\begin{aligned} d(x^{(n)}, x) &= \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{d_j(x_j^{(n)}, x_j)}{1 + d_j(x_j^{(n)}, x_j)} \\ &= \sum_{j=1}^k \frac{1}{2^j} \frac{d_j(x_j^{(n)}, x_j)}{1 + d_j(x_j^{(n)}, x_j)} + \sum_{j=k+1}^{\infty} \frac{1}{2^j} \frac{d_j(x_j, y_j)}{1 + d_j(x_j, y_j)} \\ &< \sum_{j=1}^k \frac{1}{2^j} \frac{\varepsilon}{2} + \sum_{j=k+1}^{\infty} \frac{1}{2^j} \\ &< \frac{\varepsilon}{2} \sum_{j=1}^{\infty} \frac{1}{2^j} + \frac{\varepsilon}{2} \sum_{j=1}^{\infty} \frac{1}{2^j} \\ &= \varepsilon \end{aligned}$$

Thus $x^{(n)} \rightarrow x$ in (X, d) .

(c). Note the following for all $x, y \in \ell^p$:

$$|x_i - y_i| \leq \left(\sum_{n=1}^{\infty} |x_n - y_n|^p \right)^{1/p} = d_p(x, y)$$

This implies the following:

$$d(x, y) = \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{|x_j - y_j|}{1 + |x_j - y_j|} \leq \sup_i |x_i - y_i| \leq d_p(x, y)$$

which shows that any sequence that converges in ℓ^p also converges in (X, d) . The converse is not true however. The sequence $x^{(n)} = e_n = (0, \dots, 0, 1, 0, \dots)$ with 1 in the n -th entry and zero elsewhere is convergent in (X, d) , since $x_i^{(n)} \rightarrow 0$ for each i . However, for any $n \neq m$, $d_p(x^{(m)}, x^{(n)}) = 2^{1/p}$ for $p \neq \infty$ and 1 if $p = \infty$, so the sequence isn't even Cauchy. $\square$

Problem 3

Let $f : \mathbb{R}_{\geq} \rightarrow \mathbb{R}$ be a continuous function s.t. $\lim_{n \rightarrow \infty} f(nx) = 0$ for all $x \in \mathbb{R}$ (here $\mathbb{R}_{\geq} := \{x \in \mathbb{R} : x \geq 0\}$). Prove that $\lim_{x \rightarrow \infty} f(x) = 0$.

Hint: apply the Baire Category theorem in the following steps:

- (a) Let $\varepsilon > 0$ and define $E_k := \{x : |f(nx)| \leq \varepsilon, \forall n \geq k\}$. Show that E_k is closed.
- (b) Show that $\mathbb{R}_{\geq} = \bigcup_k E_k$.
- (c) Conclude by the BCT that some E_k contains an interval (a, b) .
- (d) Show that $\bigcup_{n \geq k} (na, nb) \leq (T, \infty)$ for some $T > 0$.
- (e) Conclude that $\limsup_{t \rightarrow \infty} |f(t)| \leq \varepsilon$ and so $\lim_{t \rightarrow \infty} f(t) = 0$.

Proof. (a). Let $\varepsilon > 0$ and define $E_k := \{x : |f(nx)| \leq \varepsilon, \forall n \geq k\}$. For any sequence $x_m \in E_k$ that converges to $x \in \mathbb{R}$, we have $nx_m \rightarrow nx$ as $m \rightarrow \infty$ for any $n \geq k$, and since $|f(nx_m)| \leq \varepsilon$ and by continuity, we have $\varepsilon \geq \lim_{m \rightarrow \infty} |f(nx_m)| = |f(nx)|$. Hence $x \in E_k$ so E_k is closed.

(b). Given $x \in \mathbb{R}_{\geq}$, we have $\lim_{n \rightarrow \infty} f(nx) = 0$, so for some k , $|f(nx)| < \varepsilon$ whenever $n \geq k$. Thus $x \in E_k$ and so $\mathbb{R}_{\geq} \subseteq \bigcup_k E_k$. The other subset inclusion is obvious.

(c). By BCT, $\mathbb{R}_{\geq}$ being complete and a union of countably many closed sets E_k implies one of E_k 's has a non-empty interior. That is some E_k contains an interval (a, b) .

(d). Note that for a large enough n , we have $n(b - a) > a$, which implies that for all n large enough, $(na, nb) \cup ((n+1)a, (n+1)b)$ is an interval. This shows that $(T, \infty) \subset \bigcup_{n \geq k} (na, nb)$ for some $T > 0$.

(e). For each $x \in (T, \infty)$, we have $x \in (na, nb)$ for some $n \geq k$, so $ny = x$ for some $y \in (a, b) \subseteq E_k$, hence $|f(x)| = |f(ny)| \leq \varepsilon$. Thus $f(x) \rightarrow 0$ as $x \rightarrow \infty$. $\square$

Problem 4

Let $S \subseteq \ell^\infty$ be the set of bounded sequences s.t.

$$\lim_{n \rightarrow \infty} |x_n| = 0$$

if $(x_n)_n \in S$.

- (a) Prove that S is closed.
- (b) Let B be the unit ball in ℓ^∞ . Show that $S \cap B$ is a closed and bounded set but is not compact.
- (c) Fix $x \in S$. Let $\mathcal{F}$ be the set of $y \in \ell^\infty$ so that $|y_n| \leq |x_n|$ for all $n \geq 0$. Show that $\mathcal{F}$ is compact.

Proof. (a). Let $x^{(n)}$ be a sequence in S that converges to $x \in \ell^\infty$. Then $|x_k| \leq |x_k^{(n)} - x_k| + |x_k^{(n)}|$ for all k, n . For any $\varepsilon > 0$, we take $N > 0, K > 0$ such that $\|x^{(N)} - x\|_\infty < \varepsilon/2$ and $|x_k^{(N)}| < \varepsilon/2$ for all $k \geq K$. By our earlier estimate, $|x_k| < \varepsilon$, hence $x_k \rightarrow 0$ as $k \rightarrow \infty$, and thus $x \in S$.

(b). $S \cap B$ is closed as an intersection of two closed sets and clearly bounded (contained in a unit ball). Notice that the sequences $e_i = (0, \dots, 0, 1, 0, \dots)$, with 1 in the i -th entry and zeros everywhere else, is in $S \cap B$. For any $i \neq j$, we have $\|e_i - e_j\|_\infty = 1$, so no subsequence of the sequence $(e_i)_i$ is even Cauchy. Thus $S \cap B$ is not compact.

(c). Let $y^{(n)}$ be a sequence in $\mathcal{F}$ $\square$

Problem 5

Let $C([0, 1])$ be the set of continuous functions $f : [0, 1] \rightarrow \mathbb{R}$ with

$$D(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|$$

For any $f \in C([0, 1])$ and $0 < \alpha \leq 1$ define,

$$\|f\|_\alpha = \sup_x |f(x)| + \sup_{x, y} \frac{|f(x) - f(y)|}{|x - y|^\alpha}.$$

Let $S := \{f : \|f\|_\alpha \leq 1\}$. Show that S is a compact set.

Proof. By Arzela-Ascoli theorem, it suffices to show that S is closed, bounded, and equicontinuous. Boundedness follows from $\|f\|_\infty \leq \|f\|_\alpha \leq 1$ for all $f \in S$. For equicontinuity, let $\varepsilon > 0$ and $\delta = \sqrt[\alpha]{\varepsilon} > 0$. For any $x, y \in [0, 1]$ with $|x - y| < \delta$ and $f \in S$, we have

$$|f(x) - f(y)| = |x - y|^\alpha \sup_{x, y} \frac{|f(x) - f(y)|}{|x - y|^\alpha} \leq |x - y|^\alpha < \delta^\alpha = \varepsilon$$

so S is equicontinuous. It remains to show that S is closed.

Let $f_n \in S$ be a sequence that converges to f (uniformly). For fixed $x \neq y$ □

Problem 6

Let $X = [0, 1]^n$ be the unit cube in $\mathbb{R}^n$. Show that multivariate polynomials are dense in $C_b(X)$.

Proof. By the Stone-Weierstrass theorem, it suffices to show that the multivariate polynomials $\mathcal{P}$ in X form a function algebra that vanishes nowhere and separates points. Clearly, for any polynomials $f, g \in \mathcal{P}$ and constant $c \in \mathbb{R}$, we have that $f + g, cf, fg$ are also polynomials, so $\mathcal{P}$ is a function algebra. The constant function 1 is a polynomial, so $\mathcal{P}$ vanishes nowhere. For distinct points $(a_1, \dots, a_n) \neq (b_1, \dots, b_n) \in X$, take i such that $a_i \neq b_i$ and note that the polynomial $f(x_1, \dots, x_n) = x_i - a_i$ is zero at $(a_1, \dots, a_n)$ but nonzero at $(b_1, \dots, b_n)$, so $\mathcal{P}$ separates points. Hence $\mathcal{P}$ is dense. □

Problem 7

Let $(b_{ij})_{i,j=1}^\infty$ be a doubly semi-infinite matrix so that

$$s := \sup_i \sum_{j=1}^\infty |b_{ij}| < \infty$$

(a) For any $x \in \ell^\infty$ define a sequence $F(x) = (F(x)_i)_i$ by

$$F(x)_i = \sum_j b_{ij} x_j^2$$

Show that F is a continuous mapping from ℓ^∞ to ℓ^∞ .

(b) Show that there is an $r > 0$ so that for any $a \in \ell^\infty$ with $\|a\|_\infty < r$ the equation

$$x = a + F(x)$$

has a solution $x \in \ell^\infty$. Show that this solution is unique among solutions that are in some sufficiently small ball in ℓ^∞ .

Proof. (a). The series that defines $F(x)_i$ converges absolutely:

$$\sum_j |b_{ij} x_j|^2 \leq \|x\|_\infty^2 \sum_j |b_{ij}| < \infty$$

so $F(x)$ is well-defined. Fix $a \in \ell^\infty$, let $\varepsilon > 0$, and take $M > 0$ such that $M > \|a\|$. Define $\delta = \min\{1, \varepsilon/(s(2M + 1))\} > 0$, let $x \in B_a(\delta)$, and note the bound for all j :

$$|x_j + a_j| \leq |x_j - a_j| + 2|a_j| < \delta + 2\|a\|_\infty \leq 2M + 1$$

We then have for all i ,

$$\begin{aligned}
|F(x)_i - F(a)_i| &= \left| \sum_j b_{ij}(x_j^2 - a_j^2) \right| \\
&\leq \sum_j |b_{ij}| \cdot |x_j + a_j| \cdot |x_j - a_j| \\
&\leq \left(\sup_i \sum_j |b_{ij}| \right) (2M + 1) \|x - a\|_\infty \\
&< s(2M + 1)\delta \\
&\leq \varepsilon
\end{aligned}$$

Thus $\|F(x) - F(a)\|_\infty \leq \varepsilon$, and hence F is continuous.

(b). Let $\Phi_a : X \rightarrow X, x \mapsto a + F(x)$ for $a \in \ell^\infty$. We find conditions on a that make F into a contraction. Let $a \in \ell^\infty$ with $\|a\|_\infty < r/2$. For any $x, y \in \ell^\infty$ with $\|x\|_\infty, \|y\|_\infty \leq r$, we have

$$|\Phi_a(x)_i - \Phi_a(y)_i| = |F(x)_i - F(y)_i| \leq 2sr \|x - y\|_\infty$$

so any $0 < r < 1/(2s)$ works. Let $B := \overline{B}_0(r)$, and note that if F can be restricted to a function $B \rightarrow B$, then we will have that F is a contraction by the above bound. To ensure of this, we show that $F(x)$ maps $x \in B$ to B .

$$|\Phi_a(x)_i| = |a_i + F(x)_i| \leq |a_i| + \sum_j |b_{ij}x_j^2| \leq r/2 + sr^2 < r$$

so $\|\Phi_a(x)\| \leq r$.

Now let $a \in \ell^\infty$ with $\|a\|_\infty < r/2$. Φ_a has a unique fixed point x_0 in B , which is then the unique solution to

$$x = \Phi_a(x) = a + F(x)$$

□

Problem 8

Problem 9

Let X be a Banach space and $F : X \rightarrow X$ a contraction. Define $G(x) := x + F(x)$. Show that G is a bijection with continuous inverse.

Proof. Let x_0 be the unique fixed point of F , and $k < 1$ the constant that makes F a contraction. Define $H_n : X \rightarrow X$ by $H_1(y) = y - F(x_0)$ and $H_{n+1}(y) = y - F(H_n(y))$ for $n \geq 1$. We then let $H : X \rightarrow X$ be $H(y) = \lim_{n \rightarrow \infty} H_n(y)$. To prove that H is well-defined it suffices to show that $H_n(y)$ is a Cauchy sequence. Note the following bound

$$\|H_{n+1}(y) - H_n(y)\| = \|F(H_n(y)) - F(H_{n-1}(y))\| \leq k \|H_n(y) - H_{n-1}(y)\| \leq \dots \leq k^n \|y - 2x_0\|$$

Given $\varepsilon > 0$, take $N > 0$ such that

$$\frac{k^N}{1 - k} \|y - 2x_0\| < \varepsilon$$

For any $N \leq n \leq m$, we have

$$\begin{aligned}
\|H_m(y) - H_n(y)\| &\leq \|H_m(y) - H_{m-1}(y)\| + \cdots + \|H_{n+1}(y) - H_n(y)\| \\
&\leq k^{m-1} \|y - 2x_0\| + \cdots + k^n \|y - 2x_0\| \\
&= k^n (1 + k + \cdots + k^{m-n-1}) \|y - 2x_0\| \\
&\leq k^N \sum_{i=0}^{\infty} k^i \|y - 2x_0\| \\
&= \frac{k^N}{1-k} \|y - 2x_0\| < \varepsilon
\end{aligned}$$

Hence H is well-defined. For any $x \in X$, we have

$$\begin{aligned}
G \circ H(x) &= G\left(\lim_{n \rightarrow \infty} H_n(x)\right) \\
&= \lim_{n \rightarrow \infty} H_n(x) + F(H_n(x)) \\
&= \lim_{n \rightarrow \infty} H_{n+1}(x) + F(H_n(x)) \\
&= \lim_{n \rightarrow \infty} x - F(H_n(x)) + F(H_n(x)) \\
&= x
\end{aligned}$$

thus $G \circ H = \text{id}$. Also we have the following:

$$\|F(x) - F(H_n(x + F(x)))\| \leq k \|x - H_n(x + F(x))\|$$

This allows us to show that $H \circ G = \text{id}$

$$\begin{aligned}
\|H \circ G(x) - x\| &= \lim_{n \rightarrow \infty} \|H_n(x + F(x)) - x\| \\
&= \lim_{n \rightarrow \infty} \|F(x) - F(H_{n-1}(x + F(x)))\| \\
&= \lim_{n \rightarrow \infty} \|F(x) - F(H_n(x + F(x)))\| \\
&\leq \lim_{n \rightarrow \infty} k \|x - H_{n-1}(x + F(x))\| \\
&= k \lim_{n \rightarrow \infty} \|H_n(x + F(x)) - x\| \\
&= k^m \lim_{n \rightarrow \infty} \|H_n(x + F(x)) - x\| \\
&\rightarrow 0 \text{ as } m \rightarrow \infty
\end{aligned}$$

Thus $H \circ G(x) = x$. Therefore H is the inverse of G .

Lastly we show that H is continuous. Consider a sequence $y_m \in X$ that converges to y . For all n, k , we have

$$\|H_n(y_m) - H_n(y)\| \leq \|y_m - y\| + k \|H_{n-1}(y_m) - H_{n-1}(y)\| = \cdots = (1 + k + \cdots + k^{n-1}) \|y_m - y\| \leq \frac{\|y_m - y\|}{1-k}$$

Taking $n \rightarrow \infty$ we see that $\|H(y_m) - H(y)\| \leq \|y_m - y\| / (1-k) \rightarrow 0$ as $m \rightarrow \infty$. Hence $H(y_m) \rightarrow H(y)$, and so H is continuous. $\square$